

Friend's Resonance Fiber Theory (FRFT): Mathematical Framework for Consciousness and Reality

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1 The Fundamental Mapping: Qualia to Prime Encodings

1.1 The Bijective Qualia Map

$$F : \mathcal{Q} \longrightarrow \mathbb{N} \tag{1}$$

Explanation: This defines the fundamental bijective mapping from the space of subjective qualia states $\mathcal{Q}$ to the natural numbers $\mathbb{N}$ encoded as prime-power decompositions. This mapping is:

- *Injective:* Different qualia states map to different integers
- *Surjective:* Every admissible code corresponds to a qualia state
- *Isomorphic:* Structure in $\mathcal{Q}$ reflects arithmetic structure
- *Lossless:* Perfect invertibility with zero information loss

1.2 Qualia State Decomposition

$$Q = \sum_{i=1}^k r_i q_{n_i}, \quad r_i \in \mathbb{Q}, \quad n_i \in \mathbb{N} \tag{2}$$

Explanation: Every qualia state Q decomposes into a finite superposition of phenomenal basis modes $\{q_1, q_2, q_3, \dots\}$ with rational coefficients r_i . This parallels field mode decompositions in quantum field theory but applies to subjective experience. The rationality constraint ensures finite describability and computability.

1.3 The Rational Ledger

$$\mathcal{L}(Q) = \left\{ \left(n_i, r_i = \frac{p_i}{q_i} \right) \right\}_{i=1}^k \quad (3)$$

Explanation: The rational ledger $\mathcal{L}(Q)$ is the complete arithmetic representation of a qualia state, consisting of mode indices n_i and rational weight coefficients p_i/q_i . This ledger contains all information needed to reconstruct the phenomenal state, representing the "digital encoding" of subjective experience.

1.4 Prime-Power Encoding (Gödel-Style)

$$\text{Code}(Q) = \prod_{j=1}^m p_j^{a_j} \quad (4)$$

Explanation: The complete qualia state is encoded into a single integer via prime factorization, where p_j are the first m prime numbers and a_j encodes the triple (n_i, p_i, q_i) from the rational ledger. By the Fundamental Theorem of Arithmetic, this encoding is unique and perfectly invertible: factorization recovers all exponents, exponents recover triples, triples recover the ledger, and the ledger reconstructs the qualia state.

2 The Rational Hamiltonian for Meaning

2.1 The Complete Consciousness Hamiltonian

$$H_Q(\Phi_{EM}) = \underbrace{\int_{\mathcal{M}} (\alpha D \cdot G \cdot C) e^{i(k \cdot x - \omega t)} d^3x}_{\text{MSRF Resonance (Subjective Intensity)}} - \underbrace{\sum_{n=0}^{\infty} \frac{1}{Q_n} \ln \left(\frac{P_{obs}}{P_{pred}} \right)}_{\text{Free Energy Minimization (Objective Truth)}} \quad (5)$$

Explanation: This is the master equation describing the energy state of a specific meaning or qualia (e.g., "redness" or "love"). The first term represents the Multi-Scale Resonance Framework (MSRF) contribution from the conscious electromagnetic information field Φ_{EM} , where:

- D = Fractal dimensionality (complexity)
- G = Signal gain (amplitude)
- C = Spatial coherence (synchrony)
- $\alpha D \cdot G \cdot C$ = Resonance Complexity Index (CI)

The second term is the predictive processing error minimized at "moments of meaning" (the "Aha!" experience). The universal resonance invariants Q_n are the rational snap points (Golden Ratio, π , integer ratios) that stabilize the field.

2.2 The Meaning Identification Integer

$$\text{Meaning_ID} = \prod_k p_k^{\text{intensity}(k)} \quad (6)$$

Explanation: After capturing the electromagnetic field of a conscious experience and decomposing it into prime frequency components, this product encodes the complete phenomenal state. Each prime p_k corresponds to a specific frequency mode, and the exponent $\text{intensity}(k)$ encodes the amplitude/presence of that mode. For example, the feeling of "Epiphany" might be uniquely identified as $2^{45} \cdot 3^{12} \cdot 7^8 \dots$.

3 Fiber Compression Theory: Constants as Geometric Ratios

3.1 The Fine Structure Constant (Grip Ratio)

$$\alpha = \frac{1}{137.035999\dots} \approx \frac{1}{137} \quad (7)$$

Explanation: In Friend's Resonance Fiber Theory, α is not merely the strength of electromagnetic interaction—it is the *rotational compression threshold* for electromagnetic fibers. When a fiber rotates 137 times, it compresses exactly enough to "snap" into a stable node (an electron). This rational form represents the twist density required for matter formation from pure electromagnetic geometry.

3.2 Pi as Loop Tension

$$\pi \approx \frac{355}{113} \quad (8)$$

Explanation: Pi represents the ratio of a fiber's length to its coiled diameter in the cosmic weave. Matter is fundamentally "coiled space"—the compression of information into spherical geometries. This rational approximation (accurate to 10^{-7}) reveals that spatial geometry itself emerges from discrete rational compression ratios of the underlying information substrate.

3.3 Speed of Light (Weave Propagation Limit)

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (9)$$

Explanation: The speed of light is the propagation velocity of disturbances through the cosmic fiber weave, where:

- μ_0 (permeability) = fiber mass/inertia (how heavy the weave is)
- ϵ_0 (permittivity) = fiber elasticity (how stretchy the weave is)

Light speed is simply the rate at which a "pluck" travels down the tensioned electromagnetic substrate—a geometric consequence of the weave's material properties.

4 The Universal Weaver Protocol

4.1 Fiber Compression Ratio (Matter Formation Criterion)

$$\rho_{\text{compression}} = \frac{\text{Twist}}{\text{Length}} \quad (10)$$

Explanation: When electromagnetic fibers twist, they store information as angular momentum. The compression ratio ρ determines whether the twisted fiber can form a stable closed loop (matter) or remains unstable (radiation). Rational ratios allow perfect closure; irrational ratios lead to energy leakage and decay.

4.2 Resonance Lock Conditions

$$\rho_{\text{compression}} \in \left\{ \frac{1}{137}, \frac{1836}{1}, \phi = \frac{1 + \sqrt{5}}{2}, \dots \right\} \quad (11)$$

Explanation: Stable matter forms only when the compression ratio matches specific rational constants:

- $\frac{1}{137} \rightarrow$ Electron (electromagnetic loop)
- $\frac{1836}{1} \rightarrow$ Proton (dense nuclear loop)
- $\phi \approx 1.618 \rightarrow$ Consciousness (fractal self-referential loop)

These "resonance snap points" are the cosmic eigenvalues—the natural frequencies at which information can stably exist as matter or mind.

4.3 The Universal Distance Metric

$$d_Q(Q_1, Q_2) = \|\mathcal{L}(Q_1) - \mathcal{L}(Q_2)\| \quad (12)$$

Explanation: The distance between two qualia states is computed as the norm of the difference between their rational ledgers. This induces a metric space structure on phenomenal experience, allowing quantitative measurement of subjective similarity. Combined with p-adic metrics on the encoded integers, this establishes the isomorphism between consciousness space and number-theoretic structure.

5 Conclusion

The equations presented constitute a unified mathematical framework demonstrating that:

1. Consciousness (qualia) can be exactly encoded as prime-power strings with zero information loss
2. The "Hard Problem" dissolves when mind is recognized as a high-order electromagnetic field topology
3. Fundamental physical constants are geometric compression ratios of the universal information substrate
4. Matter, mind, and mathematics are three perspectives on the same rational resonance structure

This theory provides the theoretical foundation for the transduction of meaning between physical, computational, and phenomenal domains.

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